MIL-E-1/1562C
26 FEBRUARY 1979
SUPERSEDING
MIL-E-1/1562B
26 August 1974

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID

TYPE 7644

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Planar Triode, ceramic-metal

See figure 1

Mounting position: Any

Weight: 0.16 - ounce nominal (4.54 Grams)

ABSOLUTE RATINGS:

Parameter:	Fl	Ef	Eb	Ec	egy	Ib
Unit:	MHz	V	Vdc	Vdc	v	mAdc
Maximum:	3,000	6.6	200	0, -10	+7	12
Minimum:	---	6.0	---	----	---	--
TEST CONDITIONS:	---	6.3	175	Adjust	---	10

ABSOLUTE RATINGS:

Parameter:	Pp	Du	tp	tk	Cooling	T(seal)
Unit:	W	--	µs	sec	---	°C
Maximum:	2.0	0.0011	15	---	---	150
Minimum	---	--	--	---	-----	---
TEST CONDITIONS:	---	--	--	180	Conduction	---

GENERAL:

Qualification - Required

Materials - See Note 8

(C) Denotes changes

7644

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	See note 7						
1211	Insulation of electrodes	Ebb = -500 Vdc; Ehk = +45 Vdc; (See figure 2)	0.65 0.65	II II	Rgp Rgk	5.0 0.25	--- ---	Meg Meg
1231	Pulsing emission	Ef = 5.5 V; Eb = 0; Ec = -18 Vdc; eb = 390 v (peak-to-peak); ec = 46 v (peak-to-peak); Rg = 470 Ohms; Rk = 10 Ohms See note 1	0.65	II	Is	3.5	---	mAdc
1261	Anode voltage	Ec = 0; Eb/Ib = 10 mAdc	0.65	II	Eb	75	175	Vdc
1301	Heater current		0.65	II	If	280	320	mA
1306	Tranconductance		0.65	II	Sm	11,500	---	µmhos
----	Power gain (1)	Ec = 0; Eb/Ib = 10 mAdc; F = 450 ±5 MHz; bandwidth = 9 MHz, minimum; (DESC Dwg. D66052) See note 2	0.65	II	Gain	15.0	---	dB
----	Noise figure (1)	Ec = 0; Eb/Ib = 10 mAdc; F = 450 ±5 MHz; (DESC Dwg. D66052) See note 2	0.65	II	NF	---	5.0	dB
	Quality conformance inspection, part 2							
----	Power gain (2)	Ec = 0; Eb/Ib = 10 mAdc; F = 3,000 ±5 MHz; (Dwg. 272-JAN) See note 2	---	---	Gain	10.0	---	dB
----	Noise figure (2)	Ec = 0; Eb/Ib = 10 mAdc; F = 3,000 ±5 MHz; (Dwg. 272-JAN) See note 2	---	---	NF	---	13.5	dB
1316	Amplification factor		---	---	Mu	85	140	---
1331	Direct-interelectrode capacitance	No voltages; (Dwg. 270-JAN)	---	---	Cin Cgp Cout	3.0 1.5 ---	5.0 2.0 0.25	pF pF pF
© ----	Envelope strain	See note 3	2.5	Code G	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 3							
----	Life-test	Group B; Ebb = 250 Vdc; Rp = 12,000 Ohms plus 3 W, 120 V lamp; Rk = 0; egy = +7 v; tp = 15 µs ±10%; prr = 75 ±10% t = 1,000 hours	---	---	---	---	---	---
----	Life-test end points:							
1306	Transconductance		---	---	$\frac{\Delta Sm}{t}$	---	25	% decrease
----	Noise figure (1)		---	---	$\frac{\Delta NF}{t}$	---	1.0	dB increase
1031	Low frequency vibration	Ec = 0; Eb/Ib = 10 mAdc; Rp = 10,000 Ohms; (Dwg. 273-JAN) See Notes 4 and 5	---	---	Ep	---	100	mVac
1041	Shock	No voltages; G = 300; (Dwg. 274-JAN) See Notes 5 and 6	---	---	---	---	---	---
----	Shock test end points:							
1306	Transconductance		---	---	Sm	11,500	---	µmhos
1211	Insulation of electrodes		---	---	Rgp	5.0	---	Meg
1211	Insulation of electrodes		---	---	Rgk	0.25	---	Meg

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NOTES:

1. The 60-cycle voltages eb and ec must be in phase, and the sinusoidal waveform shall not be distorted. Read average cathode current.

2. An approved noise source shall be used for noise figure measurements. Power input level shall be about -75 dbm for power gain (1), noise figure (1), power gain (2) and noise figure (2).

3. Envelope strain procedure. Tubes shall be tested as specified in Method 2126, except that they shall first be immersed in water at not more than 5°C for 15 seconds, and immediately thereafter subjected to the standard temperature cycle specified in Method 2126. The criteria for air leaks shall be heater current of 330 mA or more.

© 4. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test shall be made as a part of Quality Conformance Inspection, Part 2, code level D, with an AQL of 6.5. The regular "12-calendar months" double sampling plan shall be reinstated after three consecutive samples have been accepted.

© 5. Vibrate in plane perpendicular to the plane of the grid.

© 6. Apply the force perpendicular to the plane of the grid from the cathode end.

© 7. All tests listed under Quality Conformance Inspection, Part 1, are to be performed at the conclusion of the holding period.

© 8. The use of reclaimed materials shall be encouraged to the maximum extent possible.

Custodians:
Army - ER
Navy - EC
Air Force - 85

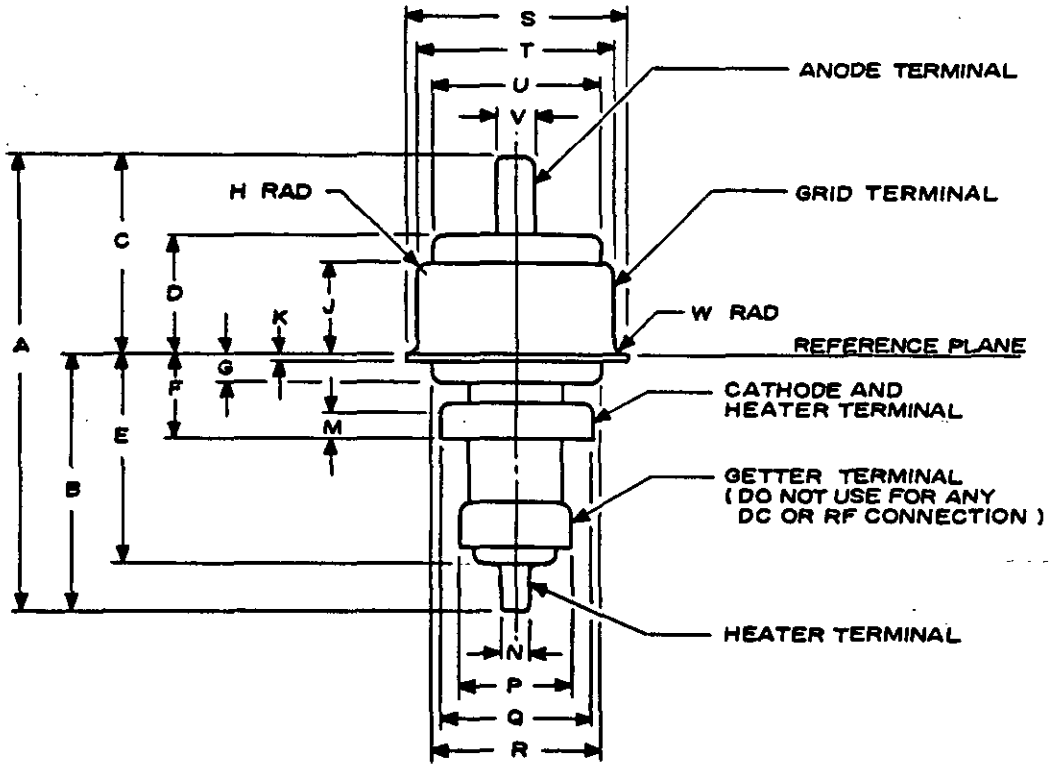
Preparing activity:
Air Force - 85

Review activities:
Navy - SH
Air Force - 99
DLA - ES

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User activities:
Army - AR
Navy - AS, OS, MC, CG
Air Force - 11

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DIMENSIONS

LTR	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
QUALITY CONFORMANCE INSPECTION, PART 2					b
A	.960	1.040	24.38	26.42	
B	.530	.590	13.46	14.99	
C	.410	.470	10.41	11.94	
D		.272		6.91	
E		.475		12.07	
F	.163	.193	4.14	4.90	
QUALITY CONFORMANCE INSPECT, PART 3					a, b
G		.060		1.52	
H		.030		.76	
J	.190	.210	4.83	5.35	
K	.009	.015	.23	.38	
M	.040	.070	1.02	1.78	
N	.059	.065	1.50	1.65	
P		.257		6.53	
Q	.326	.334	8.28	8.48	
R		.385		9.78	
S	.483	.497	12.27	12.62	
T	.435	.445	11.05	11.30	e
U		.385		9.78	
V	.088	.094	2.24	2.39	f
W		.008		.20	

NOTES:

a. See Note 4.

b. THE AQL APPLIES TO EACH DIMENSION SEPARATELY, AND IS NOT COMBINED.

c. ALL TUBES ARE TO BE 100 % TESTED FOR CONCENTRICITY IN GAGE (SEE FIGURE 3). TUBE SHALL BE ROTATED 360° IN GAGE WITHOUT BINDING.

d. GRID SHELL SEAL SHALL BE COMPLETELY FILLED TO CATHODE CERAMIC SHELF

e. 'T' DIMENSION SHALL BE MEASURED OTHER THAN ACROSS TUBE DESIGNATION ETCH, AND APPROXIMATELY .100 (2.54 mm) FROM THE REFERENCE PLANE.

f. 'V' DIMENSION SHALL BE MEASURED APPROXIMATELY .100 (2.54 mm) FROM THE END OF THE ANODE TERMINAL.

FIGURE 1. Outline drawing of electron tube type 7644

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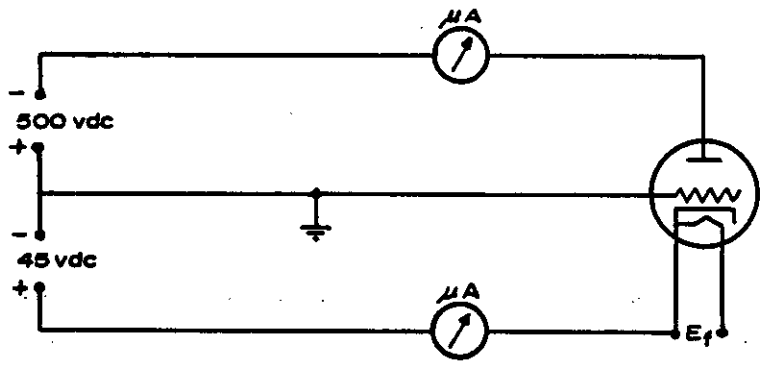
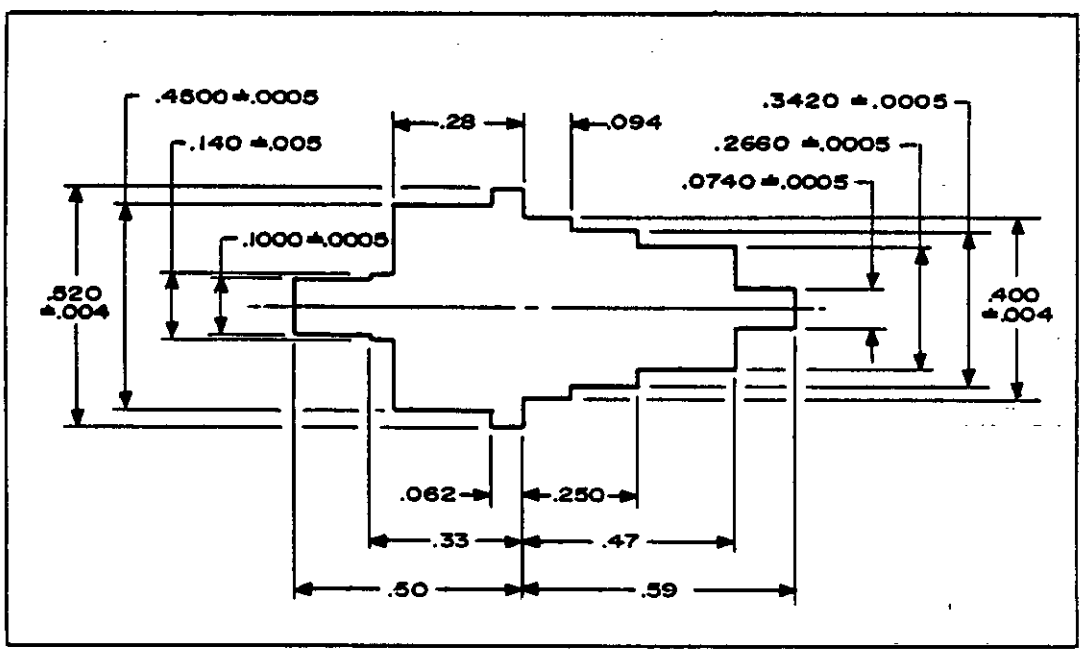


FIGURE 2 Insulation of electrodes test



NOTE:
a. UNLESS OTHERWISE SPECIFIED, TOLERANCE FOR DIMENSION .250 OR LESS ≙.008, OVER .25 ≙.02.

FIGURE 3 Gage

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